

F4 — Cycles and calendar research (2026-09-28)

This answers the deferred prompt in FUTURE_PROMPTS.md ("Deferred: cycles and calendar research"). Its start condition was met on 2026-09-28: Stages A-E of the parity roadmap are built (D2/D3 landed via #300). It is **research only**. Nothing here changes production code, and the test in Part 3 is **designed, not run**; it needs the project owner's approval.

Sources are the synthesis notes in docs/memory-bank/sources/ (no book text). Page and section references are to those notes. Throughout, "**Sources say**" is what a note records; "**Inference**" is ours.

Part 1 — What the Observation of Cycles notes say

1a. Is the solar year, and the equinox in particular, a sound anchor for Gann's time counts?

Sources say:

- Dewey separates recurrent events (discrete dated hits) from repetitive cycles (sequences), and says they need different statistics (C05, items 5–6). Anniversary and seasonal dates are recurrent events.
- Dewey's taxonomy of causes: free (internal) oscillations versus forced (external) ones. Day and night, the seasons and the tides are his named forced oscillations. A forced cycle resumes in phase after a disturbance; a free one resumes out of phase (C06, "Causes"; criterion 5).
- Halberg: a rhythm can be free-running or synchronized by an external zeitgeber such as light. Feeding can override light, and other animals can synchronize a group (C07, lesson 3). Near periods beat against the calendar year, so a component can be prominent in one stretch and vanish in another (C07, lesson 5).
- Tomes' solar review ties several market-adjacent periods to solar rotation harmonics (C03). His oil note ties a 5.5-year period to half the sunspot average (C02). Both are cross-series corroboration, not market evidence, and GSPS excludes solar timing inputs (AGENTS.md "Astrology").

What the sources do not say: none of the seven notes discusses the equinox, the solstices or the solar-year start as an anchor for price cycles. No note compares a Jan 1 count with an equinox count.

Inference:

- If markets carry a seasonal (annual) component at all, it is a forced oscillation in Dewey's sense: the season drives crops, cash flows, tax and fiscal calendars. A forced cycle keeps the forcer's phase. The forcer's phase is the astronomical year, so counting from a fixed point of that year (the equinox) is **no worse than, and in principle more faithful than**, counting from Jan 1, which is a civil convention. For equity markets, though, several forcers are civil (fiscal quarters, tax dates, January flows). Those would favour calendar anchors. The sources give no way to rank the two for US equities. That is an empirical question, which is what Part 3's test is for.
- Halberg's synchronizer point argues that the anchor that matters is whatever actually entrains the market, and that can differ by market (grains to the growing season, equities perhaps to the fiscal calendar). Gann's own evidence is mostly commodities and U.S. Steel.

Verdict: the sources make the equinox anchor **plausible as a forced-cycle anchor**, not proven. They supply no evidence that it beats Jan 1 for equities.

1b. Why could Gann's dated forecasts miss even if the underlying cycles are real?

Sources say (each is a named mechanism):

1. **Average versus specific occurrence** (Tomes, C01): a cycle can be real on average while no single interval equals the average; an alignment averaging 3.446 years actually recurs at 3.244 or 4.542 years. A forecast that plants a flag on the average date misses most specific occurrences.
2. **Crests are rarely on time** (Dewey, C06 finding 3): most crests of the 41-month cycle fell within ± 3 –4 months, about $\pm 9\%$ of the period, and extra crests appear. A dated forecast with a window of days is far tighter than that.
3. **Distortion and phase shifts** (C06 criterion 5): an internal (free) cycle resumes out of phase after a shock. Gann's own premise, "cycles repeat because human nature does not change", is an internal cause in Dewey's terms (C06 finding 4). The 1940 forecast was written in the middle of the largest shock of the century (the fall of France); an internal cycle would be expected to lose its phase there.
4. **Beats and regime dependence** (Halberg, C07 lessons 4–5): components come and go; a result that held in one regime failed in the next. A cycle can be real and still absent for a stretch.
5. **Inversion by phase** (Halberg, "feedsideways", C07 lesson 6): the same input can stimulate, do nothing, or inhibit, depending on the phase it arrives at. The equivalent in a forecast is a turn date that brings a low where a high was called.
6. **Repetition count** (C05, C06 criterion 3): long cycles have few repetitions in any record, so their "phase" is poorly known.
7. **Persistence after discovery** (C06 criterion 7): Benner's integer-year pig-iron rule worked for decades and then slipped a year every five waves, because the true period was fractional. A method built on round numbers of years drifts.

Inference: all seven apply to Gann's undisclosed Master Time Factor forecasts. They explain misses **without** requiring the cycles to be unreal, but they equally explain the misses if the cycles are unreal. The 1940 record (A07) therefore cannot, on its own, say which. It does say that dated, narrow-window forecasting from these cycles is unreliable, which is the practical point for GSPS: time counts belong as **windows with tolerance**, measured, never as dated calls.

1c. Do the sources support or undermine the 100-year and 20-year cycles?

Sources say:

- **100-year:** Gann's evidence is analogy (1840–50 against 1940–50 in A07; transport history in A03; crops in A08). The notes record one repetition in US history. Dewey's repetition-count criterion fails it by his own standard (C06 criterion 3; the coin-toss argument, C05). Tomes himself declined a 30–36-year claim for having too few repetitions (C02), which is the standard applied correctly.
- **20-year:** Gann's primary claims are the "investors' panic about every 20 years" with seven listed instances (A04, A05) and "the most important time cycle is the 20-year cycle" (Master Course). The notes check it: the intervals are 20, 16, 20, 3, 18 and 6–17 years (A05, three-question note), so **repetition count clears but regularity of timing fails**. On the lattice side, Tomes' 2×3 table has 17.79 and 22.x-year entries and a 22-year modulator of the 41-month stock cycle (Lane 1950, C01), and Dewey's 18.2-year cycle shows synchrony across 16 series (C06 criterion 12). None of those is 20.0 years.

Inference:

- The **100-year cycle is undermined** as a cycle claim: one repetition is anecdote, whatever its cause.
- The **20-year cycle is partly supported**: the literature has robust cycles near it (17.75–18.2 and ~22 years) and Gann's list has enough repetitions, but its intervals are irregular, and "20" may be a rounding of an 18- or 22-year family (Benner's lesson). It is best treated as a **window of roughly**

16-22 years, not a fixed 240 months, and measured before anything leans on it.

Part 2 — Do Tomes' harmonic ratios support Gann's $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ division of the year?

Sources say:

- Tomes' corrected theory (C04) finds the strongest harmonics on a "main line" with about **one ratio of 3 for every two to three ratios of 2**, and higher primes rare. Dewey's 1967 "Cycle Frequency Harmonics" (C01) generates most common cycles by $\times/\div 2$ and 3.
- Gann ranks the year: anniversary $> \frac{1}{2} > \frac{1}{4}$ and $\frac{3}{4} > \frac{1}{3}$ and $\frac{2}{3} > \frac{1}{8}$ s (Master Course; A2.1 notes).

Inference:

- The **order of the top of Gann's ranking matches** the lattice: halving (ratio 2) first, then the quarter (2×2), then thirds (ratio 3).
- The lattice **does not rank the eighths below the thirds**. In the 2×3 lattice $\frac{1}{8}$ is three ratio-2 steps; $\frac{1}{3}$ is one ratio-3 step. Tomes' power estimate C(H) (ordered factorisations) is $C(8) = 4$ and $C(3) = 1$, so by that measure the eighth is reached more ways than the third. Gann ranks it lower. So Tomes supports Gann's **set** of divisions (only 2s and 3s) and his preference for halves, **not his exact order below the quarter**.
- Tomes is a mechanism argument, not a market test, and says nothing about where in the year the divisions fall. It cannot choose between Jan 1 and the equinox.
- This remains **design reasoning only** (AGENTS.md: cycle literature is never evidence on its own). Gann's ranking stays because Gann gives it; Tomes explains why it is plausible.

Part 3 — A pre-registered test (designed; do not run without the owner's approval)

Question

Do Gann's dated time windows (the eight permanent-cycle windows, and the dated windows of his 1929 and 1940 forecasts) coincide with market turning points more often than chance, and does the answer depend on whether the year is counted from **Jan 1**, from the **spring equinox**, or as a **364-day (13 $\times$ 28) year**?

What is fixed before any data is looked at

1. Windows under test.

- (W1) The eight permanent-cycle windows exactly as printed (A04): Feb 8-10, Mar 21-23, May 3-7, Jun 20-24, Aug 3-8, Sep 21-24, Nov 8-11, Dec 20-24.
- (W2) Every dated window in the 1929 forecast's curve (XX major and X minor change dates). **Data gap:** our notes record only its outline and the Sep 3 top. The dates must be transcribed from the scan as a list of dates (facts, not text) before the test, and frozen.
- (W3) Every dated window in the 1940 forecast (A07): May 25 1940; Jul 10-Aug 10 1940; Sep 1 1940; Oct 1-Nov 30 1940 ("reverses"); Nov 1-11 1940; Feb 8-15 1941; Mar 1-10 1941; May 1-25 1941.

2. Three year conventions, each mapping a window to dates in every test year:

- (Y1) calendar: the printed month and day, every year.
 - (Y2) solar: the window's position measured in days from the spring equinox of 1929 (or 1940), re-applied from each year's actual equinox date.
 - (Y3) 13×28: the window's day-of-year from the anchor year, stepped forward by 364 days per year (so it drifts about 1.24 days a year against the calendar). W2 and W3 are dated to specific years; for the conventions to differ, each is also projected to later years by its offset within its year (as Gann's permanent cycle claims to repeat).
3. **Turning points.** A turn is a completed pivot of Gann's own **weekly swing chart** (WEEKLY_SWING_CHART) on the Dow Industrials daily series, 1900–2025 (and a second series, S&P 500 from 1928, as a cross-series check). Pivots are computed once and frozen. No other turn definition is tried.
 4. **Hit rule.** A window is a hit if a weekly-chart pivot falls inside it, extended by ± 2 trading days (Gann's "around"). Fixed.
 5. **Base rate.** The share of randomly placed windows of the same lengths that catch a pivot, from 10,000 circular shifts of each convention's window set within the year (shifts preserve window count, length and spacing). The p-value is the share of shifts with at least the observed hit rate.
 6. **Multiple comparisons.** 3 window sets $\times$ 3 conventions $\times$ 2 series = 18 tests. Holm–Bonferroni at family-wise $\alpha = 0.05$.
 7. **Hold-out.** Fit nothing. The windows are Gann's, so this is already out-of-sample for any year after he published them: for W1, 1930–2025 is post-discovery; W2 and W3 are scored on their own year and then projected forward. Report **before-publication** and **after-publication** separately (Dewey criterion 7).
 8. **Phase check.** For W1, split 1930–2025 into four ~ 24 -year blocks and report the hit rate per block (Halberg's time-varying dominance; Dewey criterion 6, persistence through changed conditions).
 9. **No searching.** No other anchors, tolerances, turn definitions, series or conventions. Anything else is a new, separately registered test.

Dewey items the test would clear or not

Item	Covered?
Regularity of timing	Yes: hit rate within a fixed tolerance
Repetition count	Yes: ~ 95 years for W1
Mathematical test (10)	Yes: permutation p-value with Holm correction
Persistence after discovery (7)	Yes: post-1930 scoring of W1
Persistence through changed conditions (6)	Yes: the four blocks
Cross-series clustering (11/12)	Partly: two index series, not independent markets
Dominance, constancy of period, wave shape	No: a recurrent-event test cannot address them (C05)

What it would change

- A convention that clears the correction and holds after publication would justify anchoring the fixed calendar (and G8's seasonal counts) to it. A null result would keep the calendar as it is: context, never gating.
- Either way it is a test of the disclosed calendar, not of Gann's undisclosed Master Time Factor.

Part 4 — Should the parity roadmap change?

Yes, in four places. Proposed edits (not applied; for the owner):

1. **M4 (validate calendar and anniversary windows as recurrent events).** Replace the next step with: "Run the pre-registered test in F4_CYCLES_CALENDAR_RESEARCH.md Part 3 once approved. Until then the fixed calendar and the seasonal counts stay context only."
2. **M7 (Gann's own out-of-sample record).** Add: "Scored in F4 Part 3 as window set W3; the 1929 forecast as W2 once its dated windows are transcribed as a date list."
3. **G8 (seasonal counts from March 21, built 2026-09-28).** Add to its status: "The equinox anchor is Gann's; the sources make it plausible as a forced-cycle anchor but give no evidence it beats Jan 1 for equities (F4 1a). The F4 test decides."
4. **MAJOR_CYCLE_YEARS / the 20-year cycle row.** Add: "Treat 20 years as a ~16–22-year window: Gann's instances pass repetition count and fail regularity (A05); the literature's nearby cycles are 17.75–18.2 and ~22 years (C01, C06). The 100-year cycle stays out: one repetition (F4 1c)."

No change is proposed to the 13×28 calendar: no Gann source uses it, so it enters only as a comparison convention in the test, never as a GPS feature.

Three-question basis for this research

1. **Gann:** the windows, the forecasts and the ranking are Tier A (A04, A07, the Master Course). The Master Time Factor itself is undisclosed and is not tested here as a method, only through its dated outputs.
2. **Dewey and Tomes:** the whole design is built on Dewey's criteria 3, 6, 7 and 10 and his recurrent-event distinction; Tomes supplies the lattice argument in Part 2; Halberg supplies the time-varying and phase cautions.
3. **Hermetic: Rhythm** (whether the year's returning points carry the market's turns) and **Correspondence** (whether a count that holds in one series holds in another). **Cause and Effect** frames 1a: a forced cycle keeps its forcer's phase. Mentalism, Polarity, Vibration and Gender don't shape the design.